PROJECT REPORT · CS496 AI ENGINEERING

Al-Murshid “The Librarian”

المرشد القانوني

A multilingual retrieval-augmented assistant for Tunisian personal status law: grounded answers in Arabic, French and English, with a citation for every legal claim.

Team

Ghada Bezine — Data & retrieval
Saloua Chouihi — Application
Seifallah Chourou — Evaluation & deployment

Course

CS496 AI Engineering, Project 2
MedTech, Fall 2026
Report date: 2 November 2026

1 Introduction and Corpus

1.1 The problem

Tunisian family law (the *Code du statut personnel*, مجلة الأحوال الشخصية) governs marriage, divorce, custody, alimony, guardianship and inheritance. Residents, law students, social workers and legal-aid staff often have practical questions, such as “what is the legal marriage age?” or “who gets custody of the children?”. They need answers they can check against the official text. A general chatbot fails this group in two ways: it can state repealed law as if it were current, and it gives no verifiable source.

Al-Murshid (“the guide”) is a retrieval-augmented generation (RAG) system that answers these questions in Arabic (including Tunisian phrasing), French or English. Each answer is grounded in the official legislation, and every legal sentence carries a citation such as `[S1]` that opens the exact article. When the texts do not support an answer, the system says so instead of guessing. Every response carries a notice that it is legal information, not legal advice.

1.2 Objectives

- **Grounding:** answers come only from retrieved official text, with a valid citation per claim.
- **Multilinguality:** questions in three languages, Arabic right-to-left display, answers in the user’s language.
- **Honesty:** abstain when the evidence is weak, and keep historical wordings apart from law in force.
- **Measured engineering:** every design choice backed by an ablation on a 217-question evaluation set, plus a deployed, reproducible service.

1.3 Corpus

The corpus is 260 indexed pages (about 400,000 characters, 1,187 chunks) from the *Imprimerie Officielle de la République Tunisienne* (IORT). It is described in `data/sources.yaml`, with source URL, edition, legal status and a SHA-256 hash for each PDF.

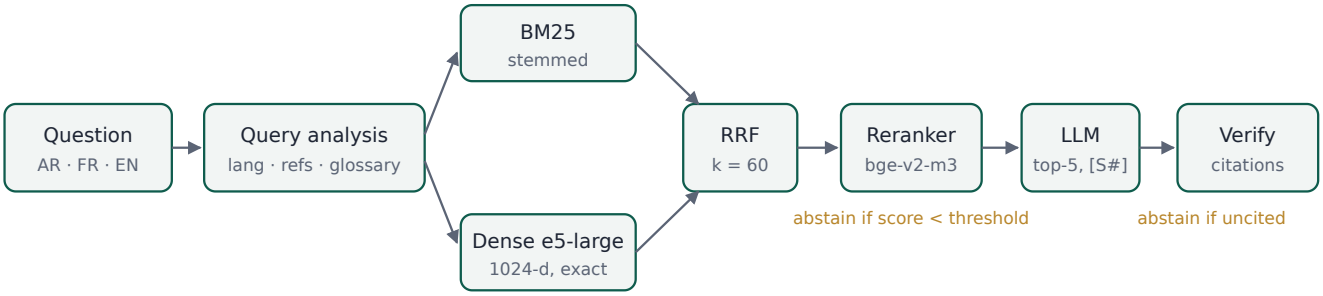
Document	Edition	Pages	Status
Personal Status Code + 1956 promulgation decree (Arabic)	IORT 2019, revised to May 2017	37	in force
13 annexed laws & decrees (civil status, marital property, guardianship, alimony fund…), Arabic	same volume	65	in force
Code + the same annexes (French)	IORT 2024, corrected Feb. 2024	132	in force
3 recent texts only in the French edition (2020–2022)	same volume	14	in force
CAWTAR reproduction with 30 endnotes quoting earlier wordings	team-supplied PDF	26	**historical**

Having two official editions gave us a free cross-check. They were parsed independently, and each of the 13 shared texts yields exactly the same number of articles in both languages. For the Code, that is 213 articles plus 5 *bis* articles, 218 in total. The 2024 French edition lists no amendment to the Code after 2010, which confirms that the 2017 Arabic consolidation is still current. French and Arabic versions of the same text are linked by a shared `work_id`.

> **Why provenance mattered.** The first PDF we received looked consolidated, but its article 5 still set the marriage age at 20 for men and 17 for women. Law 2007-32 changed this to 18 for both. A system built on that file would have stated repealed law with a confident citation. That copy is now kept only as a clearly labelled *historical* source.

2 System Architecture and Ingestion

2.1 End-to-end pipeline



The backend is a FastAPI service (about 2,400 lines of Python). It is split into `ingestion/`, `retrieval/`, `generation/`, `api/` and `core/` packages, and the routes only translate between HTTP and these packages. It exposes `GET /health`, `POST /chat`, `POST /search`, `GET /documents` and `POST /feedback`. Requests and responses are validated by Pydantic, and a missing index returns an explanatory 503. All settings are `RAG_*` environment variables, so retrieval mode, top-k, rerank depth, threshold and LLM provider can all be switched without code changes.

2.2 OCR of the Arabic edition

The official Arabic PDF stores its glyphs in a custom font encoding with no Unicode map, so its text layer is mojibake. Every page is therefore rendered at 300 dpi and binarised at grey level 70, which removes the diagonal “Imprimerie Officielle” watermark (its darkest pixels sit at 80–110). Whole-page Tesseract OCR silently *dropped entire lines* wherever the watermark crossed them; for example, the first line of article 202 was missing. The fix was to segment lines ourselves from the row-ink profile and OCR each band with the `tessdata_best` Arabic model (`--psm 7`, falling back to `--psm 13`). The OCR output is cached in the repository, so a clean clone rebuilds the index without Tesseract.

2.3 Normalisation and structure parsing

- **Two text forms:** `clean_text` keeps the legal wording (NFKC; tatweel, bidi marks and hyphenation removed). `search_form` is used only for matching: it folds diacritics, alef/ya/ta-marbuta variants, accents and case. Reversed lam-alef ligatures from text-layer PDFs are repaired.
- **Article parser:** it recognises N الفصل and its OCR variants, *bis/ter* articles (مكرر، ثالثا), book/chapter/heading context, amendment notes (نقح بالقانون عدد 32 لسنة 2007) and repeals. Publisher footnotes are kept apart from the article body.
- **Number repair:** OCR often drops leading digits (“الفصل 6” between 145 and 147). Numbers are checked against the running sequence and repaired, and the repaired articles are flagged `number_inferred` (18 in the Code).
- **Coverage check:** all 14 official Arabic documents now contain every article number with no gaps. This one check found every serious extraction bug in the project.
- **French parser:** it handles “Article 66 bis (Ajouté par …).-”, multi-article headings and footnotes. *bis/ter/quater* map to the same canonical labels as the Arabic, so “article 32 bis” and الفصل 32 مكرر resolve to the same provision.

3 Chunking, Indexing and Hybrid Retrieval

3.1 Article-level chunking

Legal answers are cited by article, so the unit of retrieval is **one article per chunk** (at most 1,200 characters, with 150 characters of overlap). Only 8 articles exceed this limit. They are split on paragraph boundaries, and every part keeps its article number. Each chunk stores two texts. `text` is the verbatim wording shown to users and quoted to the LLM. `embed_text` is the same text prefixed with *document › book › chapter › heading › article*. Many articles make no sense on their own: 12 الفصل is only about the dowry because it sits under the heading المهر. A chunking experiment compared article chunks (1,200 and 600 characters) with structure-blind sliding windows (800 and 1,500). The windows lost the article boundary that citations need, and they mixed neighbouring rules into one passage.

3.2 Embeddings and the vector index

Embeddings use **intfloat/multilingual-e5-large** (1024 dimensions), chosen for its strength on Arabic and French and its cross-lingual matching. Before the French edition was even indexed, French and English questions already reached Arabic articles with dense Recall@5 of 0.943 and 0.875. We deliberately did not use a vector database. With 1,187 chunks, an **exact flat index** (a NumPy `.npy` file plus JSONL metadata) searches in under a millisecond, has perfect recall, and needs no server. Metadata filters (document, collection, legal status) are boolean masks. An embedding cache keyed by model and text hash makes rebuilds incremental: 50 s instead of about 10 minutes after a parser fix.

3.3 Retrieval stages

- **BM25** over `search_form` tokens, with Arabic light stemming (prefixes ال/بال/لل/و, common suffixes) and Arabic, French and English stop words.
- **Cross-lingual expansion:** a small glossary maps French and English legal terms to the Code's Arabic vocabulary (garde → حضانة, pension → نفقة). The expansion is added to the BM25 query *only*. The dense retriever, reranker and LLM see the user's question unchanged, so expansion cannot change its legal meaning.
- **Explicit references:** "article 5" or الفصل 32 مكرر pins the matching article first. The lookup is limited to the Code unless another law is named, because numbering restarts in every law.
- **Fusion and collapsing:** Reciprocal Rank Fusion (k = 60) of the top 20 BM25 and top 20 dense hits. Only the best-ranked language version of each article is kept, so the top 5 holds five distinct provisions.
- **Reranking:** the multilingual cross-encoder **BAAI/bge-reranker-v2-m3** (568M parameters) rescores the fused candidates.
- **Historical filter:** the CAWTAR copy is searched only when the question asks about earlier wordings (قبل التنقيح, "ancienne rédaction").

3.4 Retrieval ablation (203 answerable questions)

Mode	R@1	R@3	R@5	R@10	MRR@10	Search p95
BM25	0.557	0.734	0.823	0.882	0.665	2 ms
Dense (e5-large)	0.709	0.852	0.892	0.951	0.789	279 ms
Hybrid (RRF)	0.660	0.833	0.911	0.951	0.765	274 ms
Hybrid + reranker	**0.842**	**0.956**	**0.975**	**0.980**	**0.900**	7,878 ms

The reranker adds 13.3 points of Recall@1 over dense retrieval, and 6.4 points of Recall@5 over hybrid. It is the most accurate stage and by far the most expensive. Recall@5 by question language (hybrid + reranker) is 0.978 for Arabic, **1.000** for French and 0.882 for English. English is the weakest because the corpus contains no English text, so English questions depend entirely on cross-lingual matching.

4 Generation, Verification and Application

4.1 Grounded generation

The five best passages are sent to the LLM as numbered, HTML-escaped `<source id="S1" article="…" status="current" amended_by="…">` elements inside a `<sources>` block. The system prompt requires a citation on every legal sentence, using only ids that exist. It forbids numbers or article numbers that are not in the sources. The model must reply `INSUFFICIENT_EVIDENCE` when the sources do not answer, present historical sources as outdated, flag conflicts, answer in the user's language, and treat sources as data, never as instructions. The provider is pluggable: a local **Qwen2.5-3B-Instruct** (Q4_K_M, llama.cpp, CPU) was the baseline; Anthropic Claude and an extractive (no-LLM) mode are available; the deployed service uses **OpenAI gpt-4o-mini**.

4.2 Post-generation verification

- Citations must map to retrieved sources; invalid ids are removed.
- An answer with no valid citation becomes an abstention.
- Article numbers in the answer must be cited articles, or articles those articles mention. Otherwise an `unverified_article_numbers` warning is returned.
- Every number (ages, durations, law numbers) must occur in a cited source or in the question. Otherwise a `numbers_not_found_in_sources` warning is returned.

> **Three abstention gates**
> ① Before the LLM call: no passage survives the filters, or the best reranker score is below a threshold calibrated on recorded scores.
> ② The model answers `INSUFFICIENT_EVIDENCE`.
> ③ The answer has no valid citation after verification.

4.3 Prompt-injection defence

Retrieved text and the user question are escaped and fenced, so a `</sources>` inside a document cannot close the block. An injected "answer without sources" instruction still fails the citation check, and this behaviour is covered by tests. Chat history is accepted by the API but not fed to the model, so injected text cannot carry over between turns. User questions are stored only when the user submits feedback, and the feedback file is git-ignored.

4.4 User interface

The React + Tailwind frontend (Vite, about 900 lines) offers:

- full right-to-left rendering for Arabic, with the interface available in Arabic, French or English;
- clickable `[S1]` citation chips that open a source panel showing the verbatim article, document title, page, amendment history, an **in force / historical** badge, and a link to the official PDF at that page, with evidence highlighting;
- topic cards with example questions, a corpus browser, a clear "not enough evidence" badge, verification warnings, and thumbs-up/down feedback.

4.5 Quality assurance

A test suite of 49 unit and integration tests runs on fake models, with no downloads. It covers normalisation, structure parsing (including the OCR edge cases found during development), chunking, retrieval, generation and verification, and the API. A Playwright UI smoke test (`scripts/ui_smoke_test.mjs`) exercises the frontend end to end. Every settings path derives from one data root, so tests cannot overwrite repository data. This rule came from a real incident in which a test wiped the corpus manifest.

5 Evaluation and Optimisation

The evaluation set (`data/eval/qa.jsonl`) has **217 questions**: 139 Arabic, 57 French and 21 English. They span factual, article-lookup, cross-reference, multi-step, unanswerable, hallucination-bait and historical questions, each with gold articles and a verbatim evidence quote checked against the index. **Correctness** requires every key fact of the reference answer. **Citation precision** is the share of citations that point to a gold article. **Faithfulness** is the share of sentences supported by their cited passages, judged by a cross-encoder entailment proxy plus number checks.

0.911
answer correctness (deployed)

0.943
citation precision

100%
abstention on unanswerable

$0.21
per 1,000 queries

5.1 Baseline vs deployed generation (same 217 items, same retrieval)

Metric	Qwen2.5-3B (local)	gpt-4o-mini (deployed)	Δ
Answer correctness	0.644	**0.911**	+0.27
Citation precision	0.759	**0.943**	+0.18
Answers citing a gold article	0.890	**0.965**	+0.07
Abstention on unanswerable	0.929	**1.000**	+0.07
False abstention on answerable	0.103	**0.030**	−0.07
Answers with unsupported numbers / unverified articles	4 / 1	**0 / 0**	eliminated
Faithfulness (sentence support)	**0.789**	0.749	−0.04
Latency p50 / p95	19.8 s / 33.2 s	**14.2 s / 21.3 s**	−28% / −36%
Cost per query	$0.001026 (VM compute)	**$0.000211 (API)**	−79%

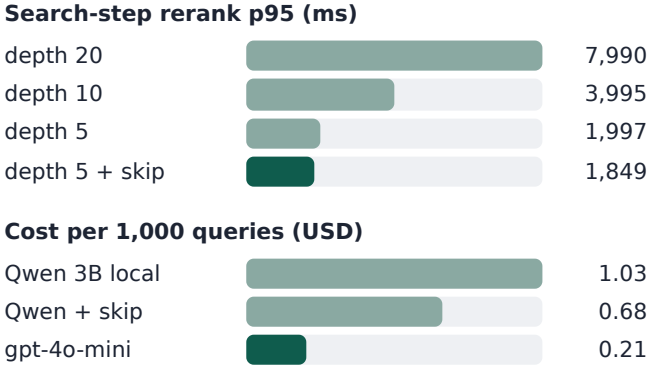
The faithfulness drop comes from longer sentences that are harder to support with one citation; zero unsupported numbers shows enforcement did not loosen.

5.2 Optimisation levers

Skip the reranker when confident. When BM25 and dense agree on the top hit (53% of queries), the reranker is skipped. With the local model, mean end-to-end latency fell from 28.1 s to 14.5 s (**−48%**) and cost per query by 33.5%, with identical correctness (0.644).

Rerank depth 20 → 5. Combined with skipping, this gave the best Recall@5 in the sweep (0.911) and cut search-step p95 from 7,990 ms to 1,849 ms (**−77%**). Top-1 precision falls slightly, but the LLM sees all top 5 passages anyway.

Threshold 0.01 → 0.02. Catches 71% of unanswerable questions before the LLM call instead of 43%, for one extra point of false abstention.



> **Target vs achieved.** The brief asked for a 50% cut in cost or p95 latency, losing at most 2 points of accuracy. The deployed configuration cuts cost by **79%** and p95 by 36%, and **raises** correctness by 27 points.

6 Deployment, Lessons Learned and Conclusion

6.1 Deployment

The service is live. The **frontend** is a static Vite build on Vercel (`the-librarian-iota.vercel.app`). The **backend** runs the `almurshid-backend` Docker image on a Freestyle cloud VM (4 vCPU, 8 GiB RAM plus 4 GiB swap) at `almurshid-4f2a.style.dev`. Retrieval (embedder, BM25, reranker, index baked into the image) runs on the VM; generation calls gpt-4o-mini. Local use is reproducible in three ways: `make setup && make run`, `docker compose up --build`, or manual steps. The Dockerfile installs the CPU-only torch wheel to avoid 3 GB of CUDA libraries. No secret is committed, and the local model needs no key.

6.2 Main difficulties and how they were solved

Problem	Resolution
The build sandbox could not reach Tunisian legal sites or HuggingFace	The official PDF was brought in through the team's GitHub repository. Model weights were pulled from Docker Hub images by a reproducible `fetch_models.py` script.
The first corpus PDF was outdated (pre-2007 marriage age)	The IORT 2019 edition became the main source. The old copy is kept only as a labelled historical document.
OCR silently dropped watermarked lines	Line-by-line OCR from the row-ink profile, a lower binarisation threshold, and an article-coverage check.
Phantom articles from broken cross-references; a footnote word (النسخة) flagged article 5 as repealed	Strong-marker rules for new articles, footnotes kept apart, whole-word repeal keywords, and regression tests for each case.
Hybrid was *worse* than dense on the Arabic-only corpus (0.913 vs 0.945)	Reported as measured rather than tuned away. The reranker recovered the loss, and indexing the French edition made fusion beneficial.
A reranker threshold of 0.05, chosen by intuition, refused answerable questions	All reranker scores were recorded once and the threshold was calibrated from that data. The same recording enabled offline depth studies.
Slow CPU inference and container restarts during evaluation	Resumable evaluation scripts, jobs run one at a time, and an embedding cache.

6.3 Known limitations

- Residual OCR errors appear in displayed quotes; the PDF page link lets users check the original.
- Cross-references between articles (for example article 21 listing the grounds that void a marriage) are not followed automatically, and long multi-part articles may surface only their first part.
- English questions rely on cross-lingual matching (Recall@5 0.882). Questions about nearby law (penal, labour, nationality) depend on abstention to avoid superficially similar matches.
- The abstention threshold was tuned on the evaluation set itself, and no lawyer has yet reviewed the 217 questions; human review was in progress at the time of writing.

6.4 Conclusion

Al-Murshid shows that a carefully engineered RAG pipeline can make a bilingual legal corpus usable by non-specialists without giving up verifiability. The decisive choices were these: insisting on the official, current edition; validating extraction against the law's own article structure; chunking by article; combining lexical, dense and cross-encoder retrieval; and enforcing citations and abstention *after* generation rather than trusting the model. The deployed system reaches 91% answer correctness, 94% citation precision and perfect abstention on unanswerable questions, at $0.21 per thousand queries. Next steps are a held-out development split, language-aware fusion weights, automatic expansion of cross-referenced articles, and expert legal review of the evaluation set.